

Novel Prep Learning Center
Math II / Enhanced Math II Placement Exam
IUSD-Aligned Readiness Diagnostic

Name: _____

Date: _____

Directions: Read each question carefully. Choose the best answer for multiple-choice questions. Show all work for graphing and free-response questions. Simplify exact answers when possible.

Teacher Use: This diagnostic answers two placement questions. Gate 1 determines whether the student has enough readiness for Math II. Gate 2 should only be applied after Gate 1 is secure and determines whether the student is ready for Enhanced Math II.

Scope	Quadratics, polynomial tools, radicals and rational exponents, complex numbers, probability, similarity, right-triangle trigonometry, circles, volume, statistics, conics, and advanced trigonometry.
Structure	55 multiple-choice questions, 4 graphing/constructed-response questions, 10 free-response questions, scoring guide, study plan, and answer key.
Placement	Use Gate 1 first. Use Gate 2 only if the student’s Math II readiness is already demonstrated.

Multiple Choice Questions

Questions 1-28 are used primarily for Math II readiness. Questions 29-55 add Enhanced Math II readiness signals, including Spring Final Review Units 6-11.

Part A — Math II Readiness

- 1 Which expression is equivalent to $\frac{a^3b^5}{a^{-1}b^2}$, assuming $a \neq 0$ and $b \neq 0$?
(A) a^2b^3 (B) a^4b^3 (C) a^3b^7 (D) $\frac{b^3}{a^4}$
- 2 Which expression shows the complete factorization of $9x^2 + 3x - 2$?
(A) $(3x + 2)(3x - 1)$ (B) $(9x - 2)(x + 1)$ (C) $(3x - 2)(3x + 1)$ (D) $(x + 2)(9x - 1)$
- 3 The vertex of $y = (x - 4)^2 - 7$ is:
(A) $(-4, -7)$ (B) $(4, -7)$ (C) $(-4, 7)$ (D) $(4, 7)$
- 4 What are the solutions of $x^2 - 5x - 14 = 0$?
(A) $x = 2, -7$ (B) $x = 7, -2$ (C) $x = -7, -2$ (D) $x = 14, -1$
- 5 The discriminant of $2x^2 + 4x + 5 = 0$ indicates that the equation has:
(A) two real rational solutions (B) one real solution (C) two real irrational solutions (D) two non-real complex solutions
- 6 Simplify $(3 + 2i)(4 - i)$.
(A) $10 + 5i$ (B) $14 + 5i$ (C) $14 + 11i$ (D) $12 - 2i$
- 7 Evaluate $27^{2/3}$.
(A) 3 (B) 6 (C) 9 (D) 18
- 8 Simplify $\sqrt{75x^4}$, assuming x is real.
(A) $5x^2\sqrt{3}$ (B) $25x^2\sqrt{3}$ (C) $5x\sqrt{3}$ (D) $75x^2$
- 9 Solve $\sqrt{x + 5} = x - 1$.
(A) $x = 1$ (B) $x = 2$ (C) $x = 4$ (D) No solution
- 10 A quadratic function has zeros -3 and 5 and opens upward. Which could be its equation?

(A) $y = (x + 3)(x - 5)$ (B) $y = -(x + 3)(x - 5)$ (C) $y = (x - 3)(x + 5)$ (D) $y = -(x - 3)(x + 5)$

- 11 What is the average rate of change of $f(x) = x^2 - 3x$ on the interval $[1, 5]$?

(A) 2 (B) 3 (C) 4 (D) 8

- 12 Which table represents a quadratic function?

(A)

x	0	1	2	3
y	2	5	8	11

(C)

x	0	1	2	3
y	2	6	18	54

(B)

x	0	1	2	3
y	1	3	7	13

(D)

x	0	1	2	3
y	4	4	4	4

- 13 Simplify $(2x^2 - 3x + 4) + (x^2 + 5x - 7)$.

(A) $3x^2 + 2x - 3$ (B) $3x^2 - 8x + 11$ (C) $2x^4 - 15x^2 - 28$ (D) $x^2 + 2x - 3$

- 14 Expand $(x - 4)(x + 6)$.

(A) $x^2 + 2x - 24$ (B) $x^2 - 2x - 24$ (C) $x^2 + 10x - 24$ (D) $x^2 + 2x + 24$

- 15 A class has 18 students who play a sport, 12 students who do not, and 10 of the athletes are girls. If one student is selected from the class, what is $P(\text{athlete})$?

(A) $\frac{3}{5}$ (B) $\frac{2}{5}$ (C) $\frac{1}{3}$ (D) $\frac{5}{9}$

- 16 If $P(A) = 0.62$, what is $P(A^c)$?

(A) 0.38 (B) 0.62 (C) 1.62 (D) -0.38

- 17 If events A and B are independent with $P(A) = 0.4$ and $P(B) = 0.7$, then $P(A \text{ and } B)$ is:

(A) 0.11 (B) 0.28 (C) 0.30 (D) 1.10

- 18 How many ways can a committee of 3 students be chosen from 8 students?

(A) 24 (B) 56 (C) 336 (D) 512

- 19 A dilation centered at the origin with scale factor 3 maps $(2, -5)$ to:

(A) $(5, -2)$ (B) $(6, -15)$ (C) $(-6, 15)$ (D) $(\frac{2}{3}, -\frac{5}{3})$

- 20 If two triangles are similar with scale factor 2 from the smaller triangle to the larger triangle, then the ratio of their areas is:

(A) 1 : 2 (B) 1 : 4 (C) 2 : 1 (D) 4 : 1

- 21 In a right triangle, an acute angle θ has opposite side 7 and hypotenuse 25. What is $\sin \theta$?

(A) $\frac{7}{25}$

(B) $\frac{24}{25}$

(C) $\frac{7}{24}$

(D) $\frac{24}{7}$

- 22 A 30° - 60° - 90° triangle has short leg 6. What is the hypotenuse?

(A) $6\sqrt{3}$

(B) 12

(C) $3\sqrt{3}$

(D) 18

- 23 The equation of a circle with center $(2, -3)$ and radius 5 is:

(A) $(x - 2)^2 + (y + 3)^2 = 25$

(C) $(x - 2)^2 + (y + 3)^2 = 5$

(B) $(x + 2)^2 + (y - 3)^2 = 25$

(D) $(x + 2)^2 + (y - 3)^2 = 5$

- 24 In a circle, an inscribed angle intercepts an arc measuring 110° . What is the measure of the inscribed angle?

(A) 55°

(B) 70°

(C) 110°

(D) 220°

- 25 A cylinder has radius 3 and height 10. What is its volume?

(A) 30π

(B) 60π

(C) 90π

(D) 300π

- 26 Which equation represents a parabola with vertex $(1, -4)$ that opens upward?

(A) $y = (x - 1)^2 - 4$

(B) $y = (x + 1)^2 - 4$

(C) $y = -(x - 1)^2 - 4$

(D) $y = (x - 1)^2 + 4$

- 27 Solve $x^2 = -49$.

(A) $x = 7$

(B) $x = -7$

(C) $x = \pm 7i$

(D) No solution

- 28 Which statement best describes the graph of $y = |x + 2| - 3$?

(A) Vertex $(-2, -3)$ and opens upward

(C) Vertex $(-2, 3)$ and opens downward

(B) Vertex $(2, -3)$ and opens upward

(D) Vertex $(2, 3)$ and opens downward

Part B — Enhanced Math II Readiness

- 29 Which piecewise function equals $2x + 1$ for $x < 0$ and x^2 for $x \geq 0$?
- (A) $f(x) = \begin{cases} x^2, & x < 0 \\ 2x + 1, & x \geq 0 \end{cases}$ (B) $f(x) = \begin{cases} 2x + 1, & x < 0 \\ x^2, & x \geq 0 \end{cases}$
(C) $f(x) = |2x + 1|$
(D) $f(x) = x^2 + 2x + 1$
- 30 The standard deviation of a data set measures:
- (A) the center of the data (C) the largest value
(B) the spread of the data from the mean (D) the middle value
- 31 A score is 84 on a normal distribution with mean 72 and standard deviation 6. What is the z -score?
- (A) -2 (B) -1 (C) 1 (D) 2
- 32 For a normal distribution, approximately what percent of data falls within one standard deviation of the mean?
- (A) 34% (B) 68% (C) 95% (D) 99.7%
- 33 A confidence interval for a population proportion is $(0.41, 0.53)$. Its point estimate is:
- (A) 0.06 (B) 0.12 (C) 0.47 (D) 0.94
- 34 Which equation represents an ellipse centered at the origin with horizontal major axis length 10 and vertical minor axis length 6?
- (A) $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (C) $\frac{x^2}{10} + \frac{y^2}{6} = 1$
(B) $\frac{x^2}{9} + \frac{y^2}{25} = 1$ (D) $\frac{x^2}{5} + \frac{y^2}{3} = 1$
- 35 A parabola has vertex $(2, -1)$ and directrix $y = -3$. Which equation represents the parabola?
- (A) $(x - 2)^2 = 8(y + 1)$ (C) $(y + 1)^2 = 8(x - 2)$
(B) $(x - 2)^2 = -8(y + 1)$ (D) $(y + 1)^2 = -8(x - 2)$
- 36 Convert 150° to radians.
- (A) $\frac{5\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{7\pi}{6}$

- 37 Evaluate $\sin\left(\frac{3\pi}{2}\right)$.
- (A) -1 (B) 0 (C) 1 (D) $\frac{\sqrt{2}}{2}$
- 38 Evaluate $\tan(210^\circ)$.
- (A) $-\sqrt{3}$ (B) $-\frac{\sqrt{3}}{3}$ (C) $\frac{\sqrt{3}}{3}$ (D) $\sqrt{3}$
- 39 A triangle has sides $a = 7$, $b = 10$, and included angle $C = 60^\circ$. By the Law of Cosines, c^2 equals:
- (A) 49 (B) 79 (C) 149 (D) 249
- 40 In $\triangle ABC$, $A = 35^\circ$, $a = 8$, and $b = 10$. Which law would be used first to determine possible measures of angle B ?
- (A) Pythagorean Theorem (B) Law of Sines (C) Law of Cosines (D) Area formula
- 41 A sector has radius 6 and central angle $\frac{2\pi}{3}$ radians. What is its arc length?
- (A) 4π (B) 6π (C) 12π (D) 24π
- 42 A chord of a circle is perpendicular to a radius. What must be true?
- (A) The radius bisects the chord (B) The chord is a diameter (C) The chord is tangent (D) The radius is twice the chord
- 43 If two tangents are drawn from the same external point to a circle, then:
- (A) the tangent segments are congruent (B) the tangents are parallel (C) the intercepted arcs are equal to 90° (D) the tangents pass through the center
- 44 A hyperbola has center $(0,0)$, vertices $(\pm 4, 0)$, and foci $(\pm 5, 0)$. Which equation represents it?
- (A) $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (B) $\frac{y^2}{16} - \frac{x^2}{9} = 1$ (C) $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (D) $\frac{x^2}{25} - \frac{y^2}{16} = 1$
- 45 A cone has radius 4 and height 9. What is its volume?
- (A) 12π (B) 36π (C) 48π (D) 144π

- 46 The table shows activity choices.

	STEM Club	Arts Club	Total
Grade 9	18	12	30
Grade 10	10	20	30
Total	28	32	60

What is $P(\text{Arts Club} \mid \text{Grade 10})$?

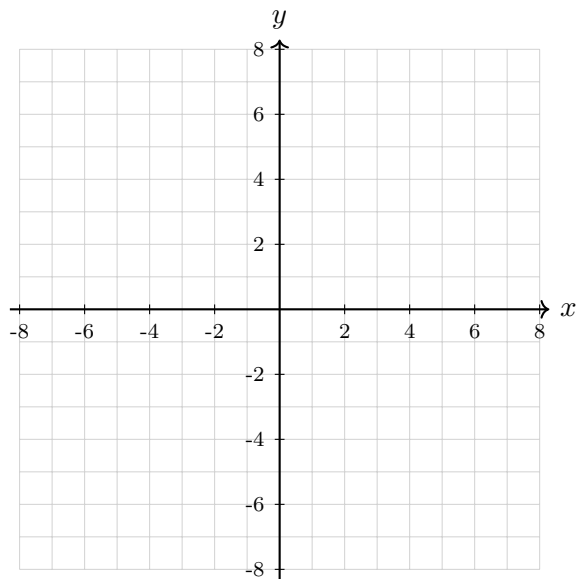
- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{5}{8}$
- 47 A club has 9 students. How many ways can it choose a president, vice president, and secretary if no student can hold more than one position?
- (A) 84 (B) 504 (C) 729 (D) 362,880
- 48 A histogram is strongly skewed right. Which statement is usually true?
- (A) The mean is greater than the median. (C) The mean equals the median.
(B) The mean is less than the median. (D) The standard deviation must be 0.
- 49 Two triangles have two pairs of corresponding angles congruent. Which similarity criterion proves the triangles similar?
- (A) SSS similarity (B) SAS similarity (C) AA similarity (D) HL congruence
- 50 The measure of one interior angle of a parallelogram is 30° less than 9 times the measure of another angle. What is the smaller angle?
- (A) 21° (B) 30° (C) 80° (D) 159°
- 51 In $\triangle ABC$, $A = 35^\circ$, $a = 12$, and $b = 18$. Which statement is true?
- (A) There is no possible triangle. (C) The triangle must be right.
(B) There is exactly one possible triangle. (D) There are two possible triangles.
- 52 Convert $\frac{7\pi}{12}$ radians to degrees.
- (A) 90° (B) 100° (C) 105° (D) 120°
- 53 Two chords intersect inside a circle. One chord has segments 6 and 8; the other has segments x and 12. Find x .
- (A) 4 (B) 6 (C) 9 (D) 16

- 54 A sector has radius 5 and central angle $\frac{4\pi}{5}$ radians. What is the area of the sector?
- (A) 5π (B) 10π (C) 20π (D) 25π
- 55 Two tangents from an external point intercept a major arc of 220° and a minor arc of 140° . What is the measure of the angle formed by the tangents?
- (A) 40° (B) 70° (C) 80° (D) 180°

Graphing and Constructed Response

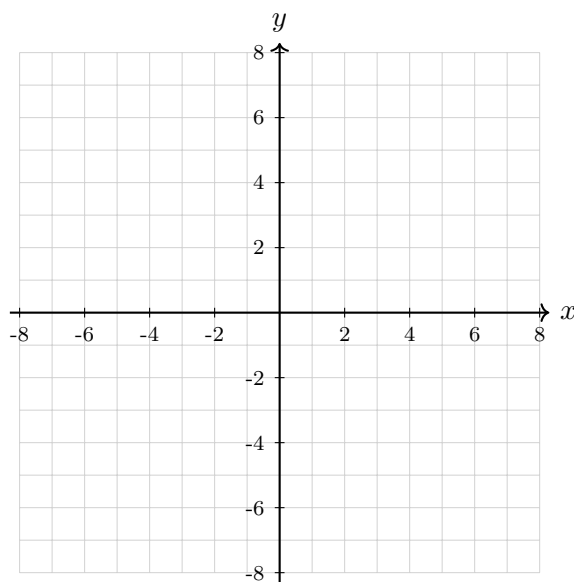
Part C — Graphs, Models, and Diagrams

- 1 Graph $y = (x - 1)^2 - 4$. Label the vertex and the x -intercepts.

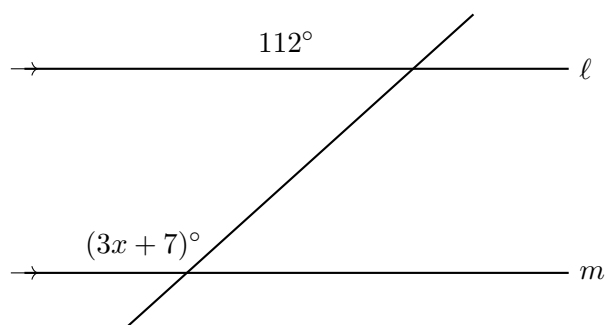


- 2 Graph the piecewise function.

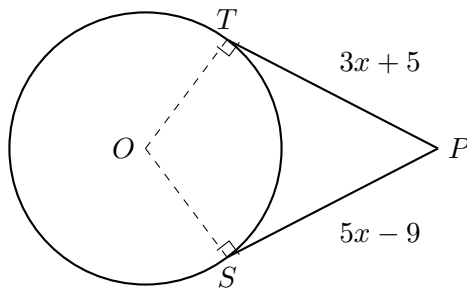
$$f(x) = \begin{cases} x + 2, & x < 1, \\ -2x + 5, & x \geq 1. \end{cases}$$



- 3 In the diagram, lines ℓ and m are parallel. Find x and justify your answer.



- 4 In circle O , PT and PS are tangent segments from the same external point P . If $PT = 3x + 5$ and $PS = 5x - 9$, find x and each tangent length.



Free Response Questions

Part D — Placement Reasoning Tasks

- 1 **Quadratic Modeling.** A ball is launched from a platform. Its height in feet after t seconds is modeled by

$$h(t) = -16t^2 + 64t + 80.$$

- (a) Find the maximum height and the time when it occurs.
- (b) Find when the ball hits the ground.
- (c) Explain what the h -intercept means in context.

- 2 **Polynomial and Complex Number Fluency.**

- (a) Factor $2x^2 - 7x - 15$ completely.
- (b) Solve $2x^2 - 7x - 15 = 0$.
- (c) Simplify $(5 - 3i) - (2 + 4i)$.

3 Radicals and Rational Exponents.

- (a) Simplify $\sqrt{108x^6}$ assuming x is real.
- (b) Rewrite $\sqrt[3]{x^5}$ using rational exponents.
- (c) Solve $\sqrt{x+3} = x+1$ and check for extraneous solutions.

4 Probability. A bag contains 5 red, 4 blue, and 3 green tiles. Two tiles are drawn without replacement.

- (a) Find the probability both tiles are blue.
- (b) Find the probability the first tile is red and the second tile is green.
- (c) Find the probability at least one of the two tiles is green.

- 5 **Similarity and Trigonometry.** A person standing 48 feet from a building measures an angle of elevation of 38° to the top of the building. The person's eye height is 5 feet.
- (a) Draw and label a right-triangle model.
 - (b) Find the height of the building to the nearest foot.
 - (c) State which trigonometric ratio you used and why.
- 6 **Enhanced Statistics.** A normally distributed test has mean 76 and standard deviation 8.
- (a) Find the z -score for a score of 92.
 - (b) Approximate the percentile for a score of 92 using the empirical rule.
 - (c) A student scored 68. Explain whether 92 or 68 is more unusual.

7 Enhanced Conics.

- (a) Write the equation of the circle with center $(-3, 4)$ and circumference 10π .
- (b) Write the equation of the ellipse centered at $(1, -2)$ with horizontal semi-major axis 5 and vertical semi-minor axis 3.
- (c) Write the equation of the parabola with vertex $(0, 1)$ and directrix $y = -1$.

8 Enhanced Triangle Trigonometry. In $\triangle ABC$, $a = 12$, $b = 15$, and $C = 42^\circ$.

- (a) Use the Law of Cosines to find side c to the nearest tenth.
- (b) Find the area of the triangle to the nearest tenth.
- (c) Explain why right-triangle trigonometry alone is not enough for this triangle.

9 **Law of Sines and the Ambiguous Case.** In $\triangle ABC$, $A = 35^\circ$, $a = 12$, and $b = 18$.

- (a) Use the Law of Sines to find all possible measures of angle B to the nearest tenth.
- (b) Find the corresponding possible measures of angle C .
- (c) Explain why this information creates the ambiguous case.

10 **Circle Relationships and Sector Area.**

- (a) Two chords intersect inside a circle. One chord has segments 5 and 12; the other has segments 6 and x . Find x .
- (b) A circle has radius 6. A sector has area 15π . Find the central angle in radians.
- (c) Find the arc length of that sector.

Teacher Scoring Guide

Suggested point values: Multiple Choice 55 points; Graphing/Constructed Response 16 points; Free Response 60 points. Total: 131 points.

IUSD alignment used for this diagnostic:

- **Math II:** quadratic and absolute value functions, polynomial expressions, quadratic equations with complex numbers, rational exponents and radicals, probability, similarity and proof, right-triangle trigonometry, circles, volume, and algebra/geometry modeling.
- **Enhanced Math II:** all Math II expectations, plus piecewise functions, stronger polynomial tools, statistics including normal distributions and standard deviation, conics, advanced triangle trigonometry, unit circle/radian fluency, and deeper circle relationships.

Two-Gate Placement Protocol

Decision Gate	Use These Items	Decision Rule
Gate 1: Math II qualification	MC 1-28, Graphing 1 and 3, FR 1-5. Maximum 66 points.	50-66: qualified for Math II. 43-49: borderline; Math II with support. Below 43: not ready for Math II without remediation.
Gate 2: Enhanced Math II qualification	Only apply if Gate 1 is passed. Use MC 29-55, Graphing 2 and 4, FR 6-10. Maximum 65 points.	50-65 and MC Part B at least 21/27: qualified for Enhanced Math II. 43-49: borderline Enhanced; targeted preparation recommended. Below 43: place in Math II or build first.

Important placement note: A student should not be placed into Enhanced Math II if the student is weak on quadratics, polynomial operations, radicals, probability, and similarity, even if the student has seen some later topics such as conics or trigonometry.

Study Plan by Diagnostic Band

Lesson length assumption: one lesson is 90 minutes. If lessons are 60 minutes, multiply the lesson count by about 1.5.

Band	Recommended Lessons	Focus
Below Math II readiness	24-32 lessons	Algebra I recovery, factoring, solving equations, function notation, graphing, exponent rules, radicals, and core geometry vocabulary. Goal: pass Gate 1 reliably.
Borderline Math II	14-20 lessons	Quadratic representations, factoring and completing the square, complex numbers, radicals, basic probability, similarity, and right-triangle trigonometry.
Math II ready, not Enhanced ready	18-26 lessons	Piecewise functions, statistics, normal models, conics, unit circle values, Law of Sines/Cosines, circle theorems, and multi-step modeling.
Borderline Enhanced	8-12 lessons	Enhanced-level mixed review: advanced trig, conics, statistics interpretation, circle proof/arc relationships, and explanation quality.
Enhanced ready	4-6 lessons	Maintenance and enrichment: proof writing, non-routine modeling, conic/trig fluency, and accuracy under timed conditions.

Module Map

Module	Lessons	When to assign
Quadratic foundations	4-6	Misses MC 2-5, 10-12, 26, 28, FR1.
Polynomial tools and complex numbers	4-6	Misses MC 1, 6, 13-14, 27, FR2.
Radicals and rational exponents	3-5	Misses MC 7-9, FR3.
Probability and counting	3-5	Misses MC 15-18, 46-47, FR4.
Similarity, proof, and geometry	5-7	Misses MC 19-20, 23-25, 49-50, Graphing 3-4, FR5.
Enhanced statistics	3-5	Misses MC 30-33, FR6.
Conics and circle equations	4-6	Misses MC 34-36, 43, FR7.
Advanced trigonometry	4-6	Misses MC 37-42, 51-52, FR8-9.
Circle relationships and sector geometry	3-5	Misses MC 42-43, 53-55, Graphing 4, FR10.

Answer Key and Selected Free Response Answers

Use this page for quick grading. Selected free-response answers show essential results; full credit should still depend on clear work and reasoning.

Multiple Choice Answer Key

Q	1	2	3	4	5	6	7	8	9
Ans	B	A	B	B	D	B	C	A	C

Q	10	11	12	13	14	15	16	17	18
Ans	A	B	B	A	A	A	A	B	B

Q	19	20	21	22	23	24	25	26	27
Ans	B	B	A	B	A	A	C	A	C

Q	28	29	30	31	32	33	34	35	36
Ans	A	B	B	D	B	C	A	A	A

Q	37	38	39	40	41	42	43	44	45
Ans	A	C	B	B	A	A	A	A	C

Q	46	47	48	49	50	51	52	53	54	55
Ans	B	B	A	C	A	D	C	A	B	A

Constructed Response Answers

- G1: Vertex $(1, -4)$; x -intercepts $(-1, 0)$ and $(3, 0)$.
- G2: Line $y = x + 2$ for $x < 1$ with open point at $(1, 3)$; line $y = -2x + 5$ for $x \geq 1$ with closed point at $(1, 3)$.
- G3: $3x + 7 = 112$, so $x = 35$ using corresponding angles.
- G4: $3x + 5 = 5x - 9$, so $x = 7$ and each tangent length is 26.

Selected Free Response Answers

- FR1: Vertex time $t = 2$; maximum height 144 ft; hits ground at $t = 5$ sec; $h(0) = 80$ is initial height.
- FR2: $(2x + 3)(x - 5)$; $x = -\frac{3}{2}, 5$; $3 - 7i$.
- FR3: $6|x^3|\sqrt{3}$; $x^{5/3}$; solution $x = 1$ only.
- FR4: $\frac{1}{11}$; $\frac{5}{44}$; $\frac{5}{11}$.
- FR5: Height $\approx 48 \tan(38^\circ) + 5 \approx 43$ ft; tangent is used because opposite and adjacent sides are involved.
- FR6: $z = 2$; about 97.5th percentile; a score of 92 is more unusual because it is 2 standard deviations above the mean, while 68 is only 1 standard deviation below the mean.
- FR7: $(x + 3)^2 + (y - 4)^2 = 25$; $\frac{(x-1)^2}{25} + \frac{(y+2)^2}{9} = 1$; $x^2 = 8(y - 1)$.

- FR8: $c \approx 10.0$; area ≈ 60.2 square units; the triangle is not given as right, so Law of Cosines/area with included angle is needed.
- FR9: $\sin B = \frac{18 \sin 35^\circ}{12}$, so $B \approx 59.4^\circ$ or 120.6° ; $C \approx 85.6^\circ$ or 24.4° .
- FR10: $x = 10$; central angle $\theta = \frac{5\pi}{6}$; arc length 5π .